Data Spaces for Data Exchange across Different Sectors and Promoting Digitalization.*

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Abstract

Electronic-Business Information Systems (EBIS), originally focused on automation, have been evolving towards value creation through the large-scale exchange of digital assets. Nowadays, data has become one of the most valuable assets for every organization. On the other hand, Data Spaces are ecosystems designed to allow multiple organizations or companies to share data in a secure, decentralized, and sovereign manner. This paper aims to describe how Data Spaces can add value within the context of EBIS and how they promote digitalization. In addition, this study will include a number of use cases where Data Spaces have been applied to facilitate the secure exchange and sharing of data across various sectors. Specifically, these include oceanography, the food industry, agriculture, and healthcare. The research will focus on presenting the basic concepts of data spaces, the benefits they can bring to EBIS, and describing use cases to demonstrate that these technologies have already helped to strengthen multiple sectors.

1 Introduction.

Since the start of the digitalization of organizational business operations, there has been a rapid shift in the approach to the global digital economy. Originally, **Electronic Business Information Systems (EBIS)** focused on automation, but this approach has evolved towards creating value through the large-scale exchange of digital assets. Nowadays, data has become one of the most valuable assets for every organization, and traditional systems may face the challenge of integrating data from multiple external sources without compromising security or integrity. Current data-sharing models often rely on centralized platforms that act as intermediaries and this approach can lead to a loss of data control and sovereignty. Therefore, the aim is to steer progress towards a model that is seamless and governed by the data providers themselves, with a focus on the scalability and sustainability of EBIS.

In this landscape, the concept of "Data Space" has emerged in recent years, with the aim of enabling secure data sharing. **Data Spaces**[1] are briefly defined as ecosystems where organizations are able to securely share data coming from different sources, ensuring data sovereignty too. Data Spaces transform the way that companies and organizations share and manage data and this may represent a change in the architectural paradigm for the next generation of E-Business. A Data Space provides a collaborative and structured environment, enabling an infrastructure where participants can share data based on common governance rules and technical standards. These principles are defined by several European initiatives such as the **International Data Spaces Association (IDSA)** [2], **Gaia-X** [3] or the **Data Spaces**

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Support Center (DSSC) [4]. The result of applying these guidelines is an infrastructure in which data from different providers can be shared with consumers in a sovereign and secure manner.

Aside from data sovereignty, the integration of data spaces also represents a step towards semantic interoperability and usage control, contributing to improve security. The Data Spaces paradigm allows digital assets to be presented in conjunction with usage policies and conditions stipulated by their owners. This permits the definition of which actions are permitted once access has been granted and how the shared data will be used, as well as defining time limits or restricting redistribution to third parties. This ability to monitor and enforce policies ensures that technical governance is an operational reality that safeguards the integrity of the asset throughout its entire lifecycle.

Throughout this paper, the intersection between the continuous evolution of EBIS and the implementation of Data Spaces will be examined. The main objective is to develop synergies between EBIS and Data Spaces, and to explore how they facilitate data exchange across different sectors while promoting digitalization. The document is organized as follows:

- **Section 2: Related Work.** This section provides an overview of the evolution of EBIS, as well as setting the context for describing how data spaces work.
- **Section 3: Data Spaces. Technical Framework.** The concepts of data spaces are explored in technical detail, and an overview of the overall architectural design is provided.
- **Section 4: Use Cases. Data Exchange in different areas.** With a greater focus on practical use cases and implementations, this section describes real-world examples from different sectors where data spaces have been used to deliver a product or service.
- **Section 5: Future Work.** This section summarizes the value that data spaces have added to EBIS based on the cases presented, and how they can continue to be applied to promote digitalization.

2 Related Work.

This section reviews the current state of the art regarding the various EBIS approaches and the emerging Data Spaces ecosystems. Key concepts such as data governance models or usage control enforcement are presented to establish a theoretical foundation.

2.1 E-Business Information Systems (EBIS).

The digitalization of business operations has been a phased evolution. Chronologically, the first technical milestone was the emergence of **Electronic Data Interchange (EDI)** [5] in the 1980s which enabled the structured exchange of commercial documents between organizations via private networks, laying the foundations for **Business-to-Business (B2B)** automation by eliminating paper from invoicing and ordering processes.

Due to the rapid growth of digitalization in recent years and the growing complexity of global markets, the integration of critical end-to-end business processes was required, but EDI focused on data transmission. **Electronic Business (e-Business)** emerged as a strategic response to the need to integrate business operations beyond the physical boundaries of traditional office processes. This paradigm comes with a major change not only in the commercial activity but also in the organization. E-Business can be defined as “the practice of performing and coordinating critical business processes such as designing products, obtaining supplies,

manufacturing, selling, fulfilling orders, and providing services through the extensive use of computer and communication technologies and computerized data". [6] This shift was driven by the need for greater agility and the realization that data, when digitized and shared, becomes the primary asset for competitive advantage.

Based on the foundational findings of the **Organization for Economic Co-operation and Development (OECD)** report[7], there are five core characteristics [6] of e-Business systems that can be synthesized:

- **The use of information and communication systems to standardize, automate, and integrate transactions.** Businesses have been using Electronic Data Interchange (EDI) to automate routine business transactions. This early stage established the first pillar of modern e-Business: the use of standardized information systems to automate and integrate transactions. From this technical foundation, e-Business emerged as a broader strategic practice, transforming simple data exchange into a systemic coordination of the entire value chain.

- **The transformation of marketplaces and the emergence of entirely new industries.** Electronic commerce is giving rise to entirely new industries and is transforming the nature of existing ones.

- **An increased connectivity and interactivity between business partners over electronic networks.** The use of a communication network provides a common, ubiquitous communications infrastructure, allowing for the seamless integration of internal business processes with those of suppliers and customers.

- **The alteration of the relative importance of time and resources in favor of digital agility.** Electronic commerce allows firms to better manage their supply chain, substituting information for physical inventory.

- **The catalytic effects of both inter- and intra-organizational transactions on the broader economy.** Electronic commerce has a catalytic effect that will stimulate more efficient ways of organizing and conducting business across the whole economy.

These five characteristics of e-Business systems may lead to various benefits, like operating more efficiently or creating more efficient business processes. Taking into consideration that e-Business is properly planned, deployed and supported, these potential benefits could be attainable in every organization. Information Systems support the fulfilment of these principles and are understood to be the foundation of e-Business.

EBIS enable companies to move away from a linear approach and start using digital 'building blocks'. It is important for organizations to carefully plan architectural frameworks for these systems, an e-business architecture planning model requires a multi-dimensional approach to align technical capabilities with business objectives, this planning framework is structured around two main components:

- **Axes that define the needs of the information architecture.** Originally[8], two axes were used to classify these needs. This evolved to three axes on which drivers of the information architecture needs of e-business firms fall.

  - *Functional Integration.* Measures the extent to which the value chain processes are digitally covered.

  - *Degree of Innovation.* Assesses the extent to which the system enables new value creation, moving from the mere digitalization of traditional tasks to the birth of native electronic markets.
  - *Structural Integration.* Evaluates the depth of the digital relationship with business partners, ranging from simple transactions to full ecosystem synchronization.

- **Generic e-business models.** They are the specific implementations, the choice of model determines how digital assets are aligned to achieve a sustainable competitive advantage. While eleven models were initially identified [8], research expanded this to twelve [9]. These models range from the operational efficiency of e-procurement to the creation of ecosystems on collaborative platforms.

Regarding the EBIS success drivers, the organization's ability to evolve through stages of maturity that validate that architecture is one of the fundamental pillars. Ensuring the right maturity level [6], and being able to measure the maturity level of **Information Systems (IS)** is about how well the IS is integrated into the organization's strategy. In order to realise the expected benefits, organizations need to turn to their EBIS and measure the maturity of those systems. In doing so, they need to identify the status of those systems with regards to their suitability to support the e-Business strategy, while also identifying required IS improvements.

2.2 Data Spaces Ecosystems.

Driven by Europe's data strategy, the **European Data Spaces initiative** [10] aims to build collaborative ecosystems based on sovereign, trusted, and interoperable frameworks. These facilitate cross-sector data sharing, allowing organizations to utilize data effectively while retaining full ownership and control.

Regarding the core benefits, **Data Spaces** [11] offer advantages that transform information management. Decentralized data exchange is facilitated, allowing each participant to retain custody of the information within their own infrastructure. This guarantees data sovereignty, ensuring that owners control and decide on how and when their data are used. Data Spaces facilitate decentralized data sharing, thereby driving value creation, fostering the development of innovative solutions and promoting the responsible data usage across the entire ecosystem.

Compared to conventional sharing methods, Data Spaces embed trust through clear governance models and technical standards, enabling a federated exchange in which each participant maintains data sovereignty, transparency, and accountability. These standards are established by the different initiatives that align efforts across Europe's data economy.

The **Data Spaces Business Alliance (DSBA)** [12] brings together four leading bodies: **Big Data Value Association (BDVA)** [13], **FIWARE Foundation** [14], **Gaia-X European Association for Data and Cloud** [15] [16] and **International Data Spaces Association (IDSA)** [17].

Under DSBA's umbrella, the **Technical Convergence Document** [18] defines a unified **Blueprint Architecture**, that establishes a shared reference architecture for building, governing, and scaling data spaces. The **Data Spaces Support Centre (DSSC)** [4], funded by the **Digital Europe Programme** [19], translates the Blueprint into practice[20]. It offers detailed blueprints and reference implementations, training materials and best-practice guides and alignment with EU policy goals and regulatory requirements.

This blueprint arranges what is needed to construct them into **Building Blocks** that contain the capabilities every data space needs, as well as best practices and specifications to implement these capabilities.[21] These Building Blocks are divided into two groups:

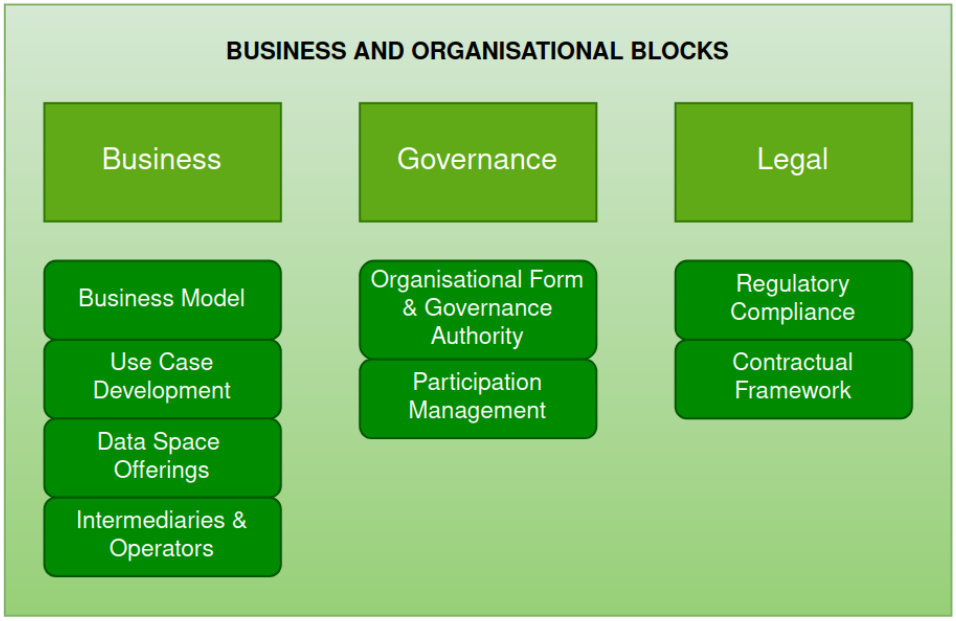


Figure 1: Business and Organizational Building Blocks.[22]

- **Business and Organizational Building Blocks.** [22] They help data space initiatives and their participants to build sustainable, compliant, and value-driven systems. As shown in Figure 1, these blocks are organized across three key pillars that shape the functionalities available to participants, guiding developers in creating robust data spaces and enabling users to effectively leverage the offerings within them.

- **Technical Building Blocks.** [23] These building blocks contemplate technical capabilities that are needed to implement data spaces. Using these building blocks makes it easier to reuse technical solutions and achieve interoperability both within and between dataspaces on a technical level. This group is also organized across three key pillars that are shown in Figure 2, regarding data interoperability, sovereignty and value creation.

The architecture of a data space is built around several core concepts that ensure the system is decentralized, secure, and interoperable. A data space is built upon the previously mentioned Technical Building Blocks, that help to create a modular, scalable, and interoperable system, enabling secure data exchange and collaboration among different entities. These building blocks are based on different tasks that are needed and integrated to create a functional data space.

Developing a data space is a complex task due to the balance that must be between business and organizational aspects, as well as technical aspects while ensuring sustainability. That is why there is a **Co-Creation Method** [24] that provides a structured, step-by-step approach to help data space participants navigate these challenges, ensuring informed decision-making and efficient collaboration. The Co-Creation Method will guide data space initiatives through the different stages of the development cycle, which is relevant for the evolution of the data space.

A member of the Data Space is a participant, a subject that is involved in the data-sharing

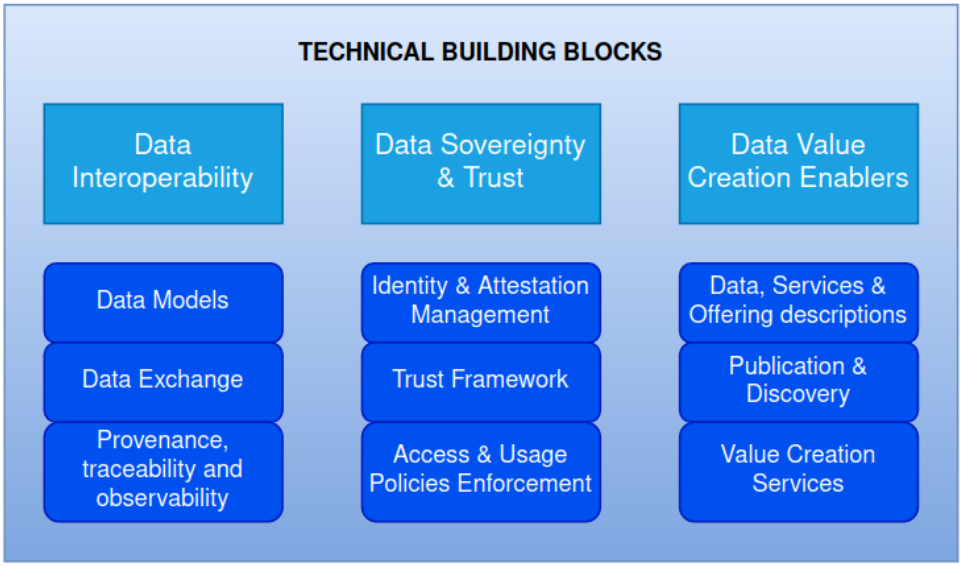


Figure 2: Technical Building Blocks.[23]

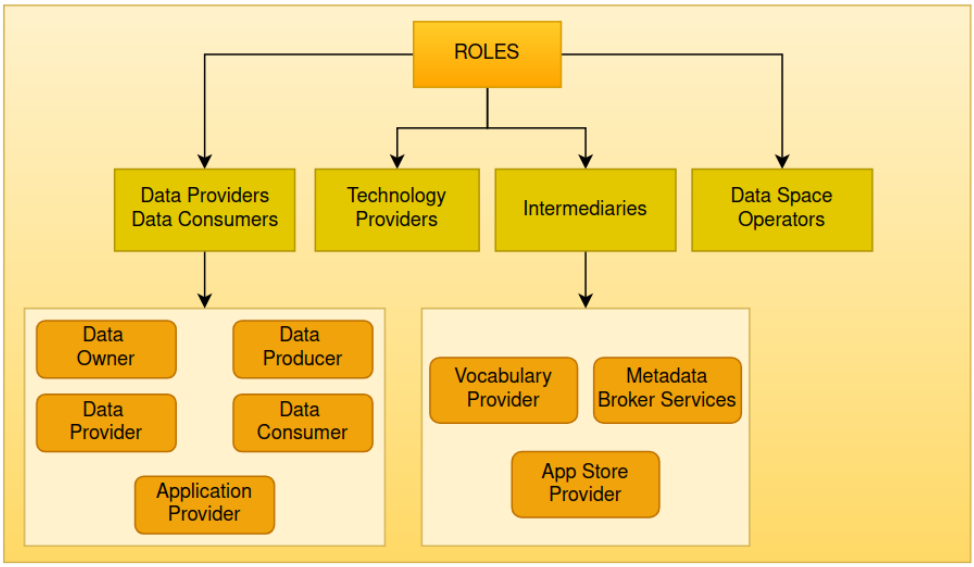


Figure 3: Roles involved in a data space.[25]

activities within the ecosystem. This participant can adopt different **roles** depending on its function and what it contributes to the Data Space, the most relevant are:

- **Data Owners.** Stipulates the conditions under which a consumer can use the data.
- **Data Providers.** Publish metadata and datasets to Catalogs [26], that is the mechanism

by which data is offered.

- **Data Consumers.** Access to Catalog, where they discover the offered data and consume datasets under agreed terms.
- **Data Intermediaries.** Offer value-added services (e.g., quality assurance, transformation).
- **Technology providers.** Supply and maintain underlying infrastructure and tools.

Participants use a **Connector** [27] [28], a fundamental component that encapsulates their role and provides a standardized interaction point among participants in the Data Space. This component is responsible for mediating secure data transfers between participants while enforcing access-control and usage-control policies according to owner stipulations. A Connector performs operations on behalf of a Participant, offers Datasets, produces Agreements and manages Dataset sharing while enforcing access-control and usage-control policies according to owner stipulations. [29] A Data Space architecture is split into a **Control Plane** [30], responsible for deciding how data are managed and processed (e.g. user identification and policy enforcement), and a **Data Plane** [31], responsible for the actual data sharing process. The connector acts as the point of convergence between the two planes: it operates in the control plane to negotiate contracts and verify identities, thereby ensuring that any movement of information in the data plane strictly complies with the agreed usage policies. A Data Space could count on different components depending on the approach adopted. In spite of this, there are some of them that are strictly necessary for a Data Space to perform its basic activity as required. One of these components and one of the main elements in the context of data spaces is the previously mentioned Connector. Many organizations have developed their own implementation of a Base Connector to seamlessly integrate within their ecosystem. Key players like the **Eclipse Foundation** [32] and the **FIWARE Foundation** [33] have been instrumental in this development by providing the reference implementations that define the modern data space landscape.

3 Data Spaces: The Next Step for EBIS.

Having introduced the basic concepts needed to understand what a data space is, this section will explore the technical implementation and how data spaces can be aligned with EBIS.

3.1 Technical Framework.

This section will examine the **Technical Building Blocks** (see Figure 2) previously mentioned in greater detail, in order to provide a foundation and context for the protocols under which Data Spaces operate. There are three groups within those technical building blocks:

- **Data Interoperability.** This block establishes the capabilities for data interoperability by defining data formats, interfaces (APIs), and semantic models that enable shared understanding within a data space. Beyond facilitating the actual exchange and sharing of data, it integrates robust mechanisms for provenance, traceability, and observability, ensuring that the entire lifecycle of data movement is documented and can be audited for regulatory compliance.
- **Data Sovereignty and Trust.** This block defines the capabilities for Data Sovereignty and Trust, establishing the essential framework for identifying participants and assets

while ensuring policy enforcement. Through Identity and Attestation Management, the system maintains the reliability and integrity of participant information, which is further validated by a Trust Framework designed to verify adherence to established rules. Ultimately, these components work together to enable the precise specification and enforcement of access and usage policies, ensuring that data remains under the control of its owner throughout its lifecycle.

- **Data Value Creation Enablers.** This block focuses on the capabilities necessary to generate tangible value within a data space by providing clear descriptions of data, services, and offerings that are understandable to all participants. These descriptions are leveraged through publication and discovery mechanisms, such as data catalogs, which allow potential users to locate and access relevant data assets efficiently. Furthermore, this block integrates value creation services that facilitate the development of additional shared tools and services, setting the necessary conditions to ensure the effective provision, delivery, and utilization of data-driven insights.

The technical realization of a Data Space moves away from centralized platforms towards a distributed architectural pattern. Based on the principles previously discussed, the framework is operationalized through the different protocols established by the European Data Spaces Initiatives.

3.1.1 Data Space Protocol.

The technical building blocks are the technical capabilities that the architecture of a Data Space requires, they will be implemented with the **Data Space Protocol (DSP)**. [34] [35]

The pillars of the DSP are the operational tools that implement the Technical Building Blocks functionalities. The DSP is the main technical support that standardizes how participants interact. At its core, the framework transitions to a Connector-based architecture. As it has been introduced previously, every participant (Providers and Consumers) operates a Connector component that serves as the sovereign gateway. These gateways communicate using a common language to handle metadata, legal constraints, and actual data transmission.

The DSP, which ensures interoperability through functional technical pillars, it is divided into three critical protocols that manage the lifecycle of a data exchange, represented on Figure 4.

- **Catalog Protocol.**[26] A **Catalog** is a collection of entries representing **Datasets** and their **Offers** that is advertised by a Provider Participant. The Catalog Protocol defines how a Catalog is requested from a Catalog Service by a Consumer using an abstract message exchange format. This is the discovery phase. Instead of a central database, the Catalog Protocol allows participants to broadcast or query available data offerings. It uses a federated approach where the Connectors exchange information about what data is available, its format, and the initial terms of use. This protocol follows the **DCAT Vocabulary** [36] and covers the Technical Building Blocks tasks that are related to **Data Value Creation Enablers**, specifically regarding data, services, and offering descriptions, as well as publication and discovery capabilities. By utilizing standardized metadata, it also supports **Data Interoperability**, ensuring that shared semantics and data models are consistently applied so that any participant can understand and interact with the discovered assets within the data space.

- **Contract Negotiation Protocol.**[37] A Consumer chooses an **Offer** from the Catalog, so that chosen asset offered has to be negotiated between the Provider and the Consumer.

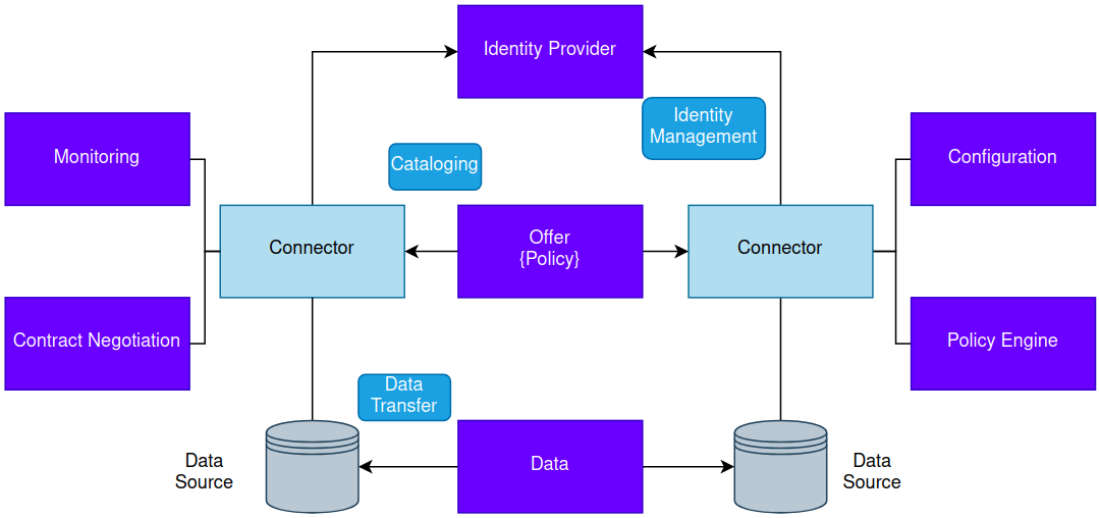


Figure 4: Overview on DSP and context. [34]

An Offer contains a concrete **Policy** associated with a specific Dataset. This protocol manages the negotiation of Usage Policies that are contained in that Offer and will result in a legally binding, digitally signed **Contract Agreement** between the **Provider** and **Consumer** that establishes the conditions under which the Data negotiated will be used. This protocol will use the **ODRL Information Model**[38] to represent Offers, Policies and Agreements. This protocol covers the Technical Building Blocks tasks that are related to **Data Sovereignty and Trust**, specifically the enforcement of Access and Usage Policies. It provides the mechanism to transition from a general Trust Framework to a specific agreement, ensuring that identity attestations and usage constraints are implicit in a digital agreement before any data transfer starts.

- **Data Transfer Protocol.**[39] Whereas the previous two protocols operate within the Connector's Control Plane (the management layer where discovery and legal agreements are handled), this protocol operates in the **Data Plane**. This protocol handles the actual movement of data bits and information. It supports various transport mechanisms (like HTTP, MQTT, or SFTP) while ensuring that the transfer is logged and compliant with the issued Contract Agreement, maintaining the sovereignty of the provider throughout the process. This protocol focuses on the Technical Building Blocks functionalities that are related to **Data Interoperability and Data Exchange**, providing the actual capabilities for the physical exchange of data. Furthermore, it implements the Provenance, Traceability, and Observability requirements by generating the necessary logs to audit the transaction and ensure it adheres to the negotiated Usage Policies Enforcement defined in the **Sovereignty and Trust block**.

3.1.2 EDC Implementation

Eclipse Dataspace Components (EDC)[32] is a set of components that enable developers to create dataspaces using a set of specific building blocks. An Identity Service that is used for managing and verifying participant identity credentials using **Decentralized**

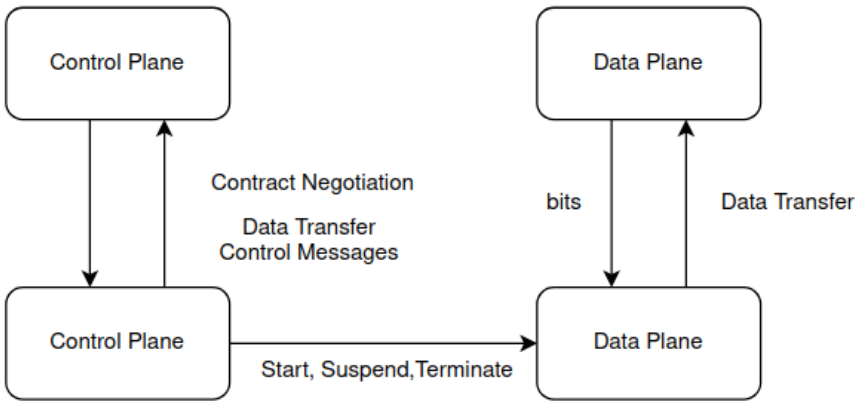


Figure 5: EDC Connector.[46]

Identifiers(DIDs)[40] and **W3C Verifiable Credentials (VCs)** [41] or **OAuth2 tokens** [42]. There is also a Catalog Service is meant for publishing assets that can be shared within the ecosystem. The Control Plane Services are in charge of automated creation and processing of data usage agreements that grant access to data, while the Data Plane and Monitoring services manage data transfers using transfer protocols.

EDC Components are standards-based and implement:

- **Dataspace Protocol Specification** [35] that define the messages exchanged in a dataspace EDC implements and builds on these asynchronous messaging patterns. how to retrieve data catalogs, conduct negotiations to create contract agreements that grant access to data, and send data over various lower-level wire protocols
- **Decentralized Claims Protocol Specification** [43], that is a set of protocols for asserting participant identities and define how trust is established between participants. The Decentralized Claims Protocol (DCP), layers on DSP.

The **Eclipse Dataspace Components (EDC) Connector** is an open-source framework designed for secure, sovereign, and policy-controlled data exchange within the data space. It enables organizations to share data while maintaining control over usage, implementing **International Data Spaces (IDS)** standards. The Connector is composed by a pair of components:

- **Control Plane**[44]**.** Controls data sharing and execute data transfer. It handles all protocol logic, including identity verification, catalog management, and contract negotiation. It is in charge of communicating with other connectors but never touches the actual raw data.
- **Data Plane**[45]**.** Is responsible for executing data transfers, sends data using specialized technology such as a messaging system or data integration platform.

The interaction between both planes is described in Figure 5. EDC components are deployed

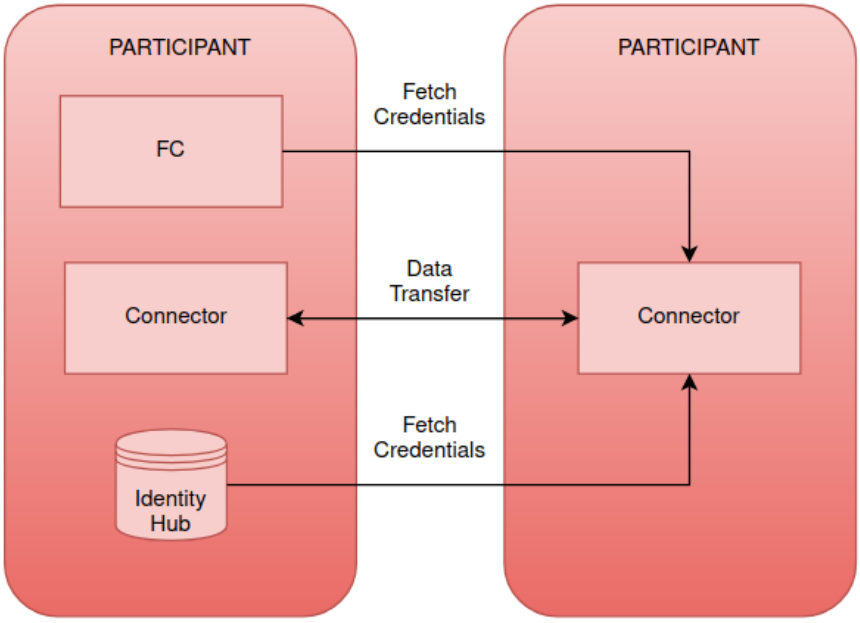


Figure 6: EDC Dataspace Context.[46]

to create a dataspace ecosystem. At its most basic level, a dataspace can be a context between two participants, as shown in Figure 6. The **Federated Catalog (FC)**[47] is responsible for crawling and caching data catalogs from other participants, it builds a local cache that can be queried or processed without resorting to complex distributed queries across multiple participants. A Connector negotiates a contract agreement for data access between two participants and manages data transfers using a data plane technology. The **Identity Hub (IH)** [48] presents verifiable credentials that a participant connector uses to determine whether it trusts and should grant data access to a counterparty. The IH securely stores and manages W3C VCs, including its presentation and issuance process.

The **EDC Minimal Viable Dataspace (MVD)**[49] sets up and runs a complete demonstration of a data space environment between two organizations.

3.2 Bridging EBIS and Data Spaces Ecosystems.

As data has become a critical asset for organizations, the current challenge for EBIS is no longer the digitalization of processes, but data governance and sovereign sharing. Data Spaces provide the necessary technical and legal layer to upgrade traditional EBIS into an interconnected ecosystem through the following core pillars:

- **Sovereignty and Decentralized Trust Framework.** Data Spaces implement a **Trust Framework** using DIDs. This allows to verify the identity of a partner that wants to participate in the data sharing process. For an EBIS, this means moving to automated, cryptographically-secured trust, where every transaction is legally binding and non-repudiable. **Decentralization** allows the EBIS to scale its network of partners rapidly, as trust is established through technical protocols. Regarding access control

and negotiation processes, Data Spaces introduce usage policies enforcement, allowing a company to share business-critical data while maintaining absolute control.

- **Interoperability. Semantic interoperability** is covered by adopting the **Catalog** concepts, EBIS data becomes discoverable and self-describing for those that can consume those data. By aligning with common data models, the EBIS ensures that business semantics are interpreted identically across different systems.

- **Value Creation.** Through the **Catalog**, an EBIS can transform its internal data into a data product available for other participants. The Data Space allows for the integration of third-party services that can process EBIS data to provide higher-level insights, such as **cross-organizational** analytics or automated compliance reporting.

- **Compliance.** The implementation of standards ensures that every exchange of business information is traceable, providing a **clear audit trail for tax and regulatory compliance**, and allowing the EBIS to monitor the status of data transfers in real-time. Organizations can monitor the status and usage of their data assets, ensuring that all external interactions remain compliant with internal governance and external regulations.

By bridging EBIS with Data Spaces, organizations move beyond the structured and limited exchange into an era of **Sovereign Intelligence**. The transition from traditional EBIS approach to Data Spaces represents a qualitative leap from the mere digitalization of processes towards a sovereign and federated data economy. By natively integrating decentralization and usage control, this architecture resolves the long-standing dilemma between open data and corporate security, enabling organizations to share critical information without compromising their competitive advantage. Data Spaces not only strengthen the resilience of systems through automated cryptographic trust, but they also set a standard for dynamic governance that transforms data into a liquid, secure and auditable strategic asset.

4 Use Cases. Data Exchange in different areas.

The University of Murcia [50] has been working with Data Spaces, drawing on EDC Components and implementing these ecosystems to apply them to Use Cases across different sectors and contexts.

4.1 Data Exchange in Oceanography.

The project called **NEREIDAS [51]** is a **Demonstration Centre** for data spaces for research and innovation in **marine sciences** in the south-western Mediterranean.

This is a managed technological platform for data spaces where various stakeholders (companies, public authorities, etc.) can carry out pilot projects and proof-of-concept tests relating to data exchange. It is an environment where data spaces are tested, validated and explained in real-world scenarios. NEREIDAS aims to demonstrate:

- **How to share data securely and independently.**

- **How to drive research and innovation through new data-driven services.**

- **How to train organizations in data technology and governance, fostering public-private collaboration.**

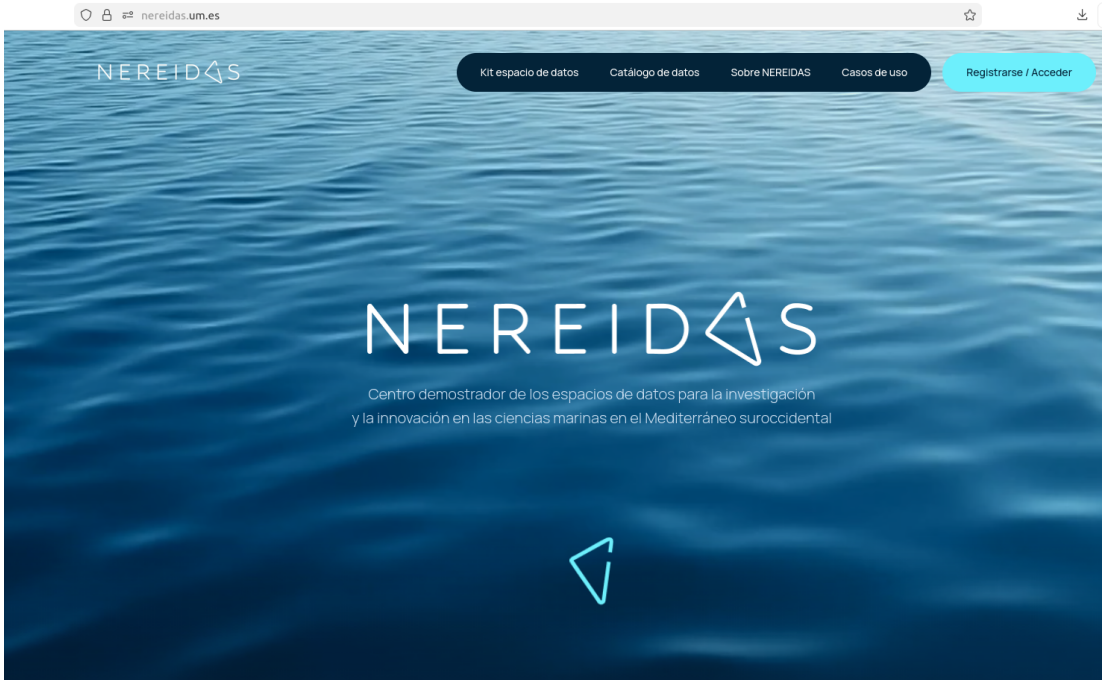


Figure 7: NEREIDAS home page.

NEREIDAS illustrate how data spaces drive **marine research and innovation in real-world scenarios**, fostering public-private collaboration, testing and validating how data ecosystems function under real-world conditions, working with the infrastructure, standards and technologies common to **European Data Spaces**. The main goal is to make this technology accessible to organizations that produce and use marine data, ensuring sovereignty, trust and security in the exchange of information.

NEREIDAS processes data on water quality, biodiversity, meteorology and catchment areas, sourced from public bodies, research institutions and the private sector. Data that is essential for understanding and managing the marine and coastal environment is treated within this project. This data is organized into four main categories:

- **Water Quality.** This includes parameters such as **temperature, salinity, chlorophyll, turbidity, dissolved oxygen, nutrients** and **pollutants**, which are key to assessing the ecological status of the marine ecosystem, preventing episodes of eutrophication and supporting decisions on coastal management or the monitoring of sensitive areas such as the Mar Menor.

- **Marine biodiversity.** Datasets on **marine species** (fish, invertebrates, underwater flora), their distribution, abundance and conservation status. These data are essential for studying trends, identifying risks, protecting habitats and designing policies for the sustainable management of the marine environment.

- **Meteorological factors.** This includes **atmospheric data that directly influence coastal and oceanic conditions, such as precipitation, air temperature, wind,**

atmospheric pressure and solar radiation. Incorporating this data helps to improve forecasting models and anticipate events relevant to sectors such as fisheries and aquaculture, tourism and port management.

- **State of the catchment area.** Information on **river channels and flows, nutrient loads, sediments, agricultural runoff** and **pollution** incidents. This data helps to link land-based impacts with changes in the marine ecosystem.

NEREIDAS takes part in the Complementary Marine Sciences Plan, within the **ThinkInAzul** [52] initiative, which connects Spain's coastal provinces to foster a shared ecosystem of data and scientific capabilities. The current situation is that marine and coastal data in Spain is fragmented, NEREIDAS was created to address this fragmentation, facilitate interoperability, and become a leading demonstration center for the secure and sovereign exchange of marine data.

4.2 Data Exchange in Food Industry.

There is an internal project where a Data Space is currently in development, called **Food Tech Specs Data Space**. The project is defined within the scope of the **Beverages Sub-sector**, in the **Agri-Food sector**. Its aim is to create and use a data platform that manages information models detailing **food product specifications, covering raw materials, semi-finished and finished products**, for a marketed food product or a specific manufactured batch.

This project addresses the need to create a use case that utilises data-sharing and processing technologies to achieve a business objective: the development and operation of a data repository that manages standardized models of food product specification information, in general terms for a **marketed food product** and specifically for a **manufactured batch**.

As shown in Figure 8, the main providers are the **primary producers and the producers in the value chain of the agri-food sector**. These would be companies that produce, process and export food products, they provide data such as the **values and technical specifications of those food products and batches, as well as a food safety alert channel**. The consumers who access the data that are being shared in the platform are the **product manufacturers or the distributors of the finished food product**, for instance, a supermarket.

A **Technical Data Sheet** for a manufactured batch is a crucial document that provides detailed information on the composition, characteristics and production processes of a specific product. In the context of beverages, this technical data sheet is essential for ensuring the quality, safety and traceability of products.

The platform aims to achieve the following objectives:

- **To provide the agri-food sector with access to a marketplace, secure and sovereign data-sharing capabilities, and a cloud infrastructure equipped with computing resources across the entire value chain, from a holistic perspective that includes data relating to the management of specifications within the agri-food sector.**

- **To provide a platform that enables the secure, reliable and controllable exchange of information. This will be aligned with current and future European data management strategies, including data traceability, access control to ensure data security and privacy, as well as compliance with European regulations.**

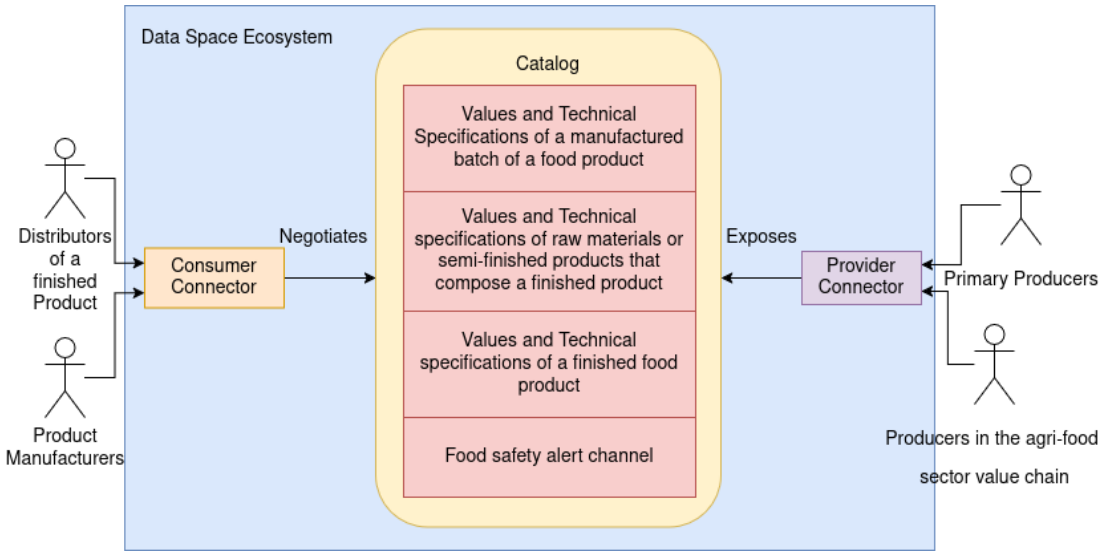


Figure 8: General overview of data sharing process in Food Tech Specs Data Space.

- **Process improvement, simplifying and streamlining the exchange of technical and quality information. The companies involved can improve the efficiency of their processes, reduce errors and response times, which in turn can lead to greater competitiveness and higher product quality.**

This data platform is unique and represents an innovative project designed to boost competitiveness within the agri-food sector, as there is currently no marketplace that facilitates the exchange of this type of information. Also, from the consumer's point of view, there is currently no platform available that allows for the centralized retrieval of information. The creation of this platform enables consumers to access detailed information on agri-food products, providing them with transparency and confidence in what they consume, fostering greater awareness of the quality and origin of the food they purchase.

4.3 Data Exchange in Healthcare and Agriculture.

The **European Open Science Cloud (EOSC)** [53] is an initiative proposed by the **European Commission** [54] and designed to provide an open environment to store and share data, encouraging scientific community to reuse research data. **EOSC Interoperability Framework (EOSC-IF)**[55] are the guidelines and technical, semantic, and legal standards that ensure consistent integration across data providers.

Funded by the European Commission, **EOSC-TITAN** [56] has the overall objective of enriching the EOSC-IF by developing a software platform solution for confidential collaboration and privacy-preserving data processing. The solution will enable access to sensitive datasets from public entities and government agencies and will be compatible by design with the EOSC-IF on the technical, semantic, organizational, and legal layers.

TITAN will leverage hardware support for **Confidential Data Processing and Data Space connectivity**, as well as novel solutions for access control policies and permissions. The project will advance the state of the art to support privacy-preserving data sharing, transparent

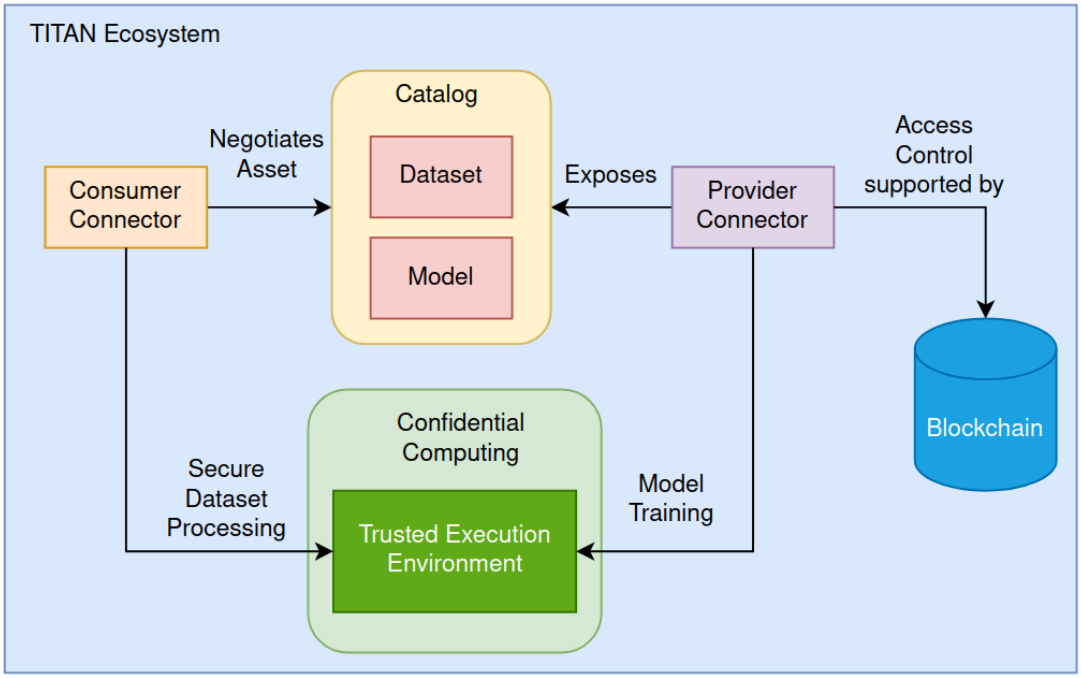


Figure 9: General overview of data sharing process and confidential computing in EOSC-TITAN.

cloud topology, auditable network security, and secure access control. The project will balance the strict security requirements with easy access for users and offer practical solutions for working with large-scale data sources, computational resources, and users.

The TITAN innovations include **Confidential Computing** and data processing based on modern efficient hardware-accelerated secure enclaves called **Trusted Execution Environments (TEEs)** [57], capable of ensuring a truly end-to-end secure data lifecycle, reliable data anonymization, and scalable and fine-grained **decentralized blockchain-based Access Control**[58].

In TITAN Platform, providers expose **datasets, models or tools** that consumers will negotiate. Access control for accessing the assets is based on a Decentralized Blockchain and the platform counts on a support for Confidential Computing where data processing and model training will be executed. Figure 9 presents a general overview of this process.

TITAN will contribute to build the **Common European Data Spaces with Government Data Zones over EOSC** [59]. These Government Zones in EOSC are sovereign, policy-governed environments designed to facilitate the secure exchange and processing of sensitive public sector information within the EOSC infrastructure. TITAN will be engaged with this target group from the government sectors by directly involving two European Regions in the project to use TITAN confidential computing and produce datasets that will become available over the EOSC, thus demonstrating Public Authorities' Government Zones in EOSC.

To promote community adoption of TITAN's open-source software, the solution will be demonstrated in two vertical cross-border scenarios in the **Agriculture and Healthcare sectors**. TITAN contributes to the **European Healthcare Data Space (EHDS)** [60] and

European Agriculture Data Space (EADS) [61], that are European Union initiatives that aims to create a secure and privacy-preserving infrastructure for the exchange of data across the EU.

TITAN services will be tested and validated supporting Confidential Computing for privacy-preserving multi-party **Machine Learning (ML)** approaches, using real-world medical data and algorithms, as well as available agriculture data from both private and public sources from the Aragón (Spain) and Veneto (Italy) regions. This will improve the usability, security, and applicability of the platform.

The pilots carried out in the project focus on real use cases in the **Agriculture and Healthcare domains**, in terms of demonstrating the general functioning of the platform and highlighting the technical solutions developed throughout the project. This diversity of scenarios makes it possible to put in common the problems of various areas in the face of privacy, security, data sharing and data exploitation.

5 Future Work

The architectural transition from traditional EBIS to Data Spaces is just the beginning of a broader digital transformation. As organizations move beyond simple connectivity and automated processes, future research and implementation will likely focus on the role of Data Spaces as the universal framework for the next generation of digitalization. The advancements in **security, sovereignty, and data governance** provided by this paradigm offer a robust foundation, that is applicable to nearly any sector. Future work should explore how the integration of sovereignty and security approach that these technologies provide, can become standard for all EBIS, regardless of their specific business logic. By embedding governance directly into the data exchange layer, future EBIS will not only automate processes but also guarantee the asset protection and integrity by design. This security-by-default point of view will be crucial as regulatory requirements and laws become more stringent across Europe. One of the most promising avenues for future development is the ability to bridge disparate domains. Data Spaces may provide the unique opportunity to relate sectors that have historically operated separately. Finally, future research will investigate the scalability of **Decentralized Trust Frameworks** and enforce the identity processes that are related to signatures and non-repudiation. As thousands of EBIS connect through a federated network, the focus will shift toward optimizing automated discovery and semantic mapping between different sector-specific data models. The goal is to create a truly liquid data economy where information flows securely and purposefully across any boundary, fueled by the legal certainty and technical sovereignty that Data Spaces provide.

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